
Post-Disaster Building Damage Assessment via a Two-Stage Deep Learning Pipeline

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Abstract

This project proposes a two-stage deep learning pipeline for post-disaster building damage assessment using satellite imagery. The system first employs U-Net to perform pixel-level segmentation and localize building regions of interest, and then applies ResNet to classify the damage level of each identified region. By decoupling localization from classification, the pipeline improves interpretability and allows independent evaluation of each module. This framework provides a practical baseline for satellite image-based disaster response, and can be further extended with change detection, attention mechanisms, or multi-task learning strategies.

1 Introduction

Natural disasters such as hurricanes, earthquakes, floods, and wildfires can cause severe damage to buildings and urban infrastructure. After a disaster occurs, rapid and accurate building damage assessment is crucial for emergency response, rescue planning, resource allocation, and post-disaster recovery. Traditional damage assessment often relies on field surveys or manual interpretation of aerial and satellite images, which can be time-consuming, labor-intensive, and difficult to scale when the affected area is large. Therefore, using machine learning methods to automatically analyze satellite imagery has become an important and practical research direction.

In this project, we focus on post-disaster building damage assessment using pre- and post-disaster satellite images. Given satellite images collected before and after a disaster, the goal is to identify building regions and classify the damage level of each building, such as no damage, minor damage, major damage, or destroyed. This task is more challenging than standard image classification because the model needs to understand not only the visual appearance of buildings, but also the structural changes caused by disasters. For example, damaged buildings may show roof collapse, boundary deformation, texture changes, or partial disappearance, while surrounding background regions such as roads, trees, shadows, and debris may introduce additional noise.

From a machine learning perspective, this problem involves both localization and classification. The model first needs to determine where the buildings are located in the image, and then predict the corresponding damage category based on visual evidence. Directly applying a classification model to the entire image may be suboptimal, since irrelevant background information can distract the model and reduce interpretability. To address this issue, we propose a two-stage vision pipeline based on deep convolutional neural networks. First, a segmentation model is used to extract building-related regions from the satellite image. Then, a classification model is applied to the segmented or cropped building regions to predict damage levels.

Specifically, we plan to use U-Net as the segmentation module and ResNet as the classification module. U-Net is suitable for pixel-level localization because its encoder-decoder architecture and skip connections help preserve both semantic context and spatial details. ResNet is selected for

damage classification because its residual connections improve the training stability of deep CNNs and allow the model to learn discriminative visual features. By combining these two models, our system separates the task into two interpretable steps: identifying the target building regions and assessing their damage severity. This design provides a clear baseline for satellite image-based disaster damage assessment and can be further extended with change detection, feature difference modeling, or attention mechanisms in later experiments.

2 Background and Proposed Method

In recent years, convolutional neural networks (CNNs) have become important foundational models in the field of computer vision. Compared with traditional methods that rely on manually designed features, CNNs can automatically learn multi-level visual representations directly from image data. Shallow convolutional layers usually capture low-level features such as edges, textures, and colors; middle layers gradually combine local shapes and structures; and deeper layers further form semantic object representations. Therefore, CNNs have been widely used in tasks such as image classification, object detection, semantic segmentation, and medical image analysis. Classic models such as AlexNet, VGG, ResNet, and U-Net have promoted the development of CNNs in visual tasks and have provided important backbones for more complex vision models.

For this project, directly feeding the entire raw image into a classification network may not be the optimal choice. The original image may contain complex backgrounds, irrelevant regions, or noisy structures. During training, the classification model may be affected by non-target regions, which can reduce classification stability and interpretability. Therefore, this project initially adopts a two-stage visual pipeline based on **segmentation-then-classification**. First, U-Net is used to perform pixel-level segmentation of the target regions in the image, generating a segmentation mask or extracting regions of interest. Then, the segmented results are fed into a ResNet classification network to further determine the target category. This design decouples the two questions of “where the target is” and “what the target is,” making the model structure clearer and allowing the segmentation quality and classification performance to be evaluated separately in later experiments.

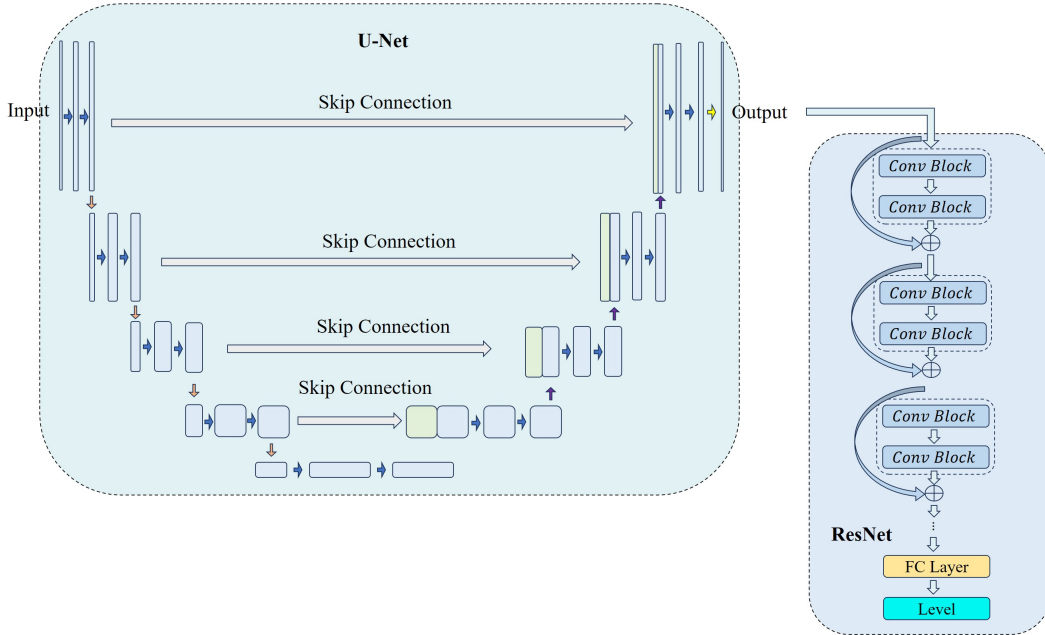


Figure 1: Overview of the proposed two-stage pipeline: U-Net for segmentation and ResNet for damage classification.

2.1 Segmentation Module: U-Net

In the segmentation stage, this project plans to use U-Net ? to extract target regions. U-Net is a classic CNN architecture for image segmentation tasks and was originally mainly used for medical image segmentation. Unlike image classification, which outputs only a single class label, segmentation requires classifying each pixel. Therefore, the model needs to understand global context while also preserving precise spatial location information.

U-Net has an overall symmetric U-shaped structure, consisting of a left-side contracting path and a right-side expanding path. The left-side encoder repeatedly applies 3×3 convolution + ReLU and 2×2 max pooling to gradually reduce the resolution of feature maps and increase the number of channels, so as to extract contextual semantic information. The right-side decoder gradually restores the spatial resolution through up-convolution and finally generates a segmentation mask.

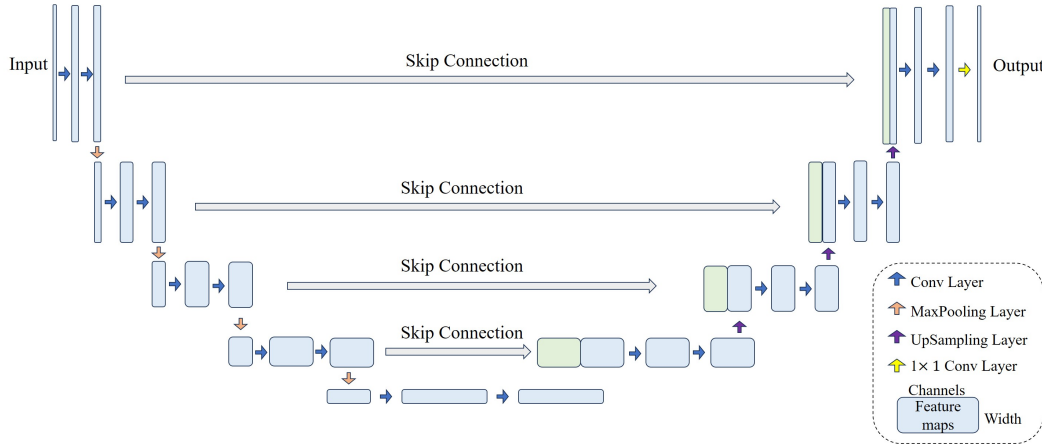


Figure 2: Architecture of U-Net.

A key advantage of U-Net lies in its skip connection design. Feature maps at the same scale from the encoder and decoder are concatenated together, allowing the decoder to use not only the semantic information from deep features, but also the edge, texture, and location information from shallow features. This design enables U-Net to balance contextual modeling and spatial detail recovery, making it suitable for segmentation tasks where object boundaries are important, background interference is strong, or labeled data is relatively limited.

In this project, U-Net is mainly used to answer the question of “where the target region is.” Specifically, it generates a target-region mask from the original image or helps crop the region of interest for the subsequent classification stage.

2.2 Classification Module: ResNet

In the classification stage, this project plans to further feed the segmentation results produced by U-Net into a ResNet classification network ?. Specifically, the segmentation mask can be applied to the original image to preserve the target region while suppressing the background. Alternatively, the target region of interest (ROI) can be cropped according to the mask, resized, and then fed into ResNet. This allows the classification model to focus more on the target region itself rather than being distracted by background information.

ResNet is a classic model for image classification in deep CNNs. Traditional CNNs aim to improve feature representation by increasing network depth. However, the ResNet paper points out that when plain networks become too deep, they may suffer from the degradation problem, where the training error increases instead of decreasing. This is not simply caused by overfitting. The core goal of ResNet is to alleviate the optimization difficulty of deep networks, allowing the model to truly benefit from increased depth.

ResNet still follows a typical classification network pipeline. The input image first passes through convolution and pooling layers to extract local features. Then, through multiple convolutional stages,

the spatial resolution is gradually reduced while the number of channels increases. Finally, global average pooling and a fully connected layer are used to output class probabilities.

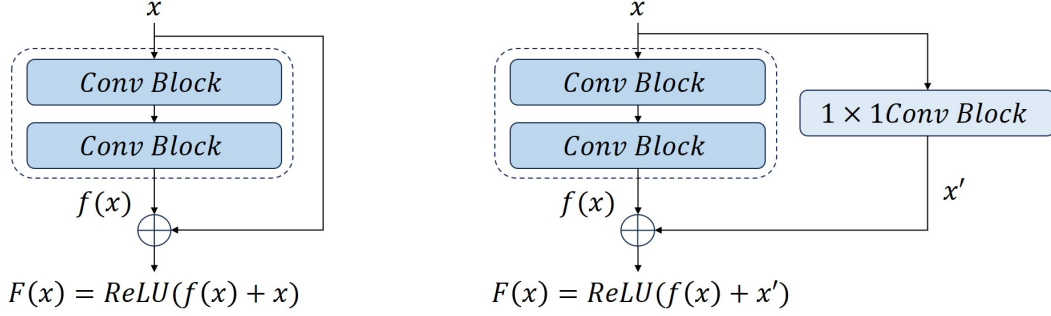


Figure 3: Architecture of Residual Block

The core innovation of ResNet is the residual connection, also known as the shortcut connection. Instead of directly learning the target mapping $H(x)$, ResNet reformulates it as:

$$F(x) = f(x) + x, \quad (1)$$

where $f(x)$ represents the residual mapping. In this way, the model only needs to learn the correction relative to the input x , rather than fitting the entire mapping from scratch. In ResNet-18, a BasicBlock usually consists of two 3×3 convolutional layers, and the input x is added to the output of the main branch through element-wise addition via the shortcut connection.

This design introduces almost no extra computational cost, but significantly improves the training stability of deep networks. As a result, ResNet can effectively extract discriminative features ranging from local textures to high-level semantic patterns. In this project, ResNet is mainly used to answer the question of “what category the target belongs to.”

3 Conclusion

In conclusion, this project proposes a two-stage deep learning pipeline for post-disaster building damage assessment using satellite imagery. The proposed method first uses U-Net to localize and extract target regions, and then applies ResNet to classify the damage level based on the segmented or cropped regions. By separating localization and classification, the system becomes more interpretable and allows the segmentation and classification modules to be evaluated independently. This framework provides a practical baseline for disaster-related visual analysis, and it can be further improved by incorporating pre- and post-disaster change modeling, attention mechanisms, or multi-task learning in future experiments.

References

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